THEALON AI ENGINEERING CHALLENGE

Podcast Companion: Research & Engineering Findings

A clear, practical review of technical choices, reliability mechanisms, and real-world benchmarks

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Live App: podcast-companion.onrender.com

Code: GitHub Public Repository

1. Why We Built This (User Value)

Long podcasts are packed with valuable information, but remembering and finding specific points later is frustrating. Standard AI chatbots usually produce generic summaries that you cannot verify.

Who benefits most:

- **Students & Learners:** Want to quickly review core ideas and test what they learned.
- **Busy Listeners:** Need to know in 2 minutes whether an 80-minute episode is worth hearing.
- **Readers & Accessibility:** People who retain information better through text than audio.
- **Podcast Creators:** Need chapter markers and quiz questions to engage their listeners.

The Core Problem Solved: Instead of guessing, we answer: "*Where did the speaker actually say that, and can I trust it?*" Every takeaway, chapter, and quiz answer links directly to the exact words in the podcast transcript.

2. Technical Tools Compared

We tested multiple free tools to find the best combination for speed, reliability, and cost.

A. AI Text Models (Summaries, Chapters, Quiz)

Model	Key Strengths	Limitations	Our Verdict
Google Gemini Flash	**Huge context window:** Reads 10+ hours in one call. Native JSON support. Generous free tier.	Free tier can experience queue delays during peak traffic hours.	**Primary Choice** (handles long episodes).
Groq (GPT-OSS-120B)	**Super fast:** Answers in 1 to 2 seconds. Reliable JSON mode.	Free tier has an 8,000 token per minute ceiling; cannot fit huge files.	**Fast Backup** & comparison model.
Hugging Face Router	Provides access to diverse open-source models with one API key.	Free tier gives small monthly credits that can run out quickly.	**Secondary Fallback**.
Local Models (Ollama)	100% private with zero API rate limits.	Needs a powerful computer/GPU. Cannot run on free cloud servers (512MB RAM).	**Rejected** for free cloud hosting.

> **Real-World Finding:** AI models retire quickly! While developing, Groq retired `llama-3.3-70b-versatile`. Because our system makes real API calls, we caught this immediately and upgraded to `openai/gpt-oss-120b`.

B. Speech-to-Text (Audio Transcribers)

Transcriber	Key Strengths	Limitations	Our Verdict
Groq Whisper	**Real timestamps:** Second-by-second accuracy via `verbose_json`. Over 200x faster than real-time.	25 MB file size limit on free tier.	**Primary Choice** (we compress and split audio into chunks).
Gemini Audio	Handles many spoken languages well natively.	Timestamps are not dependable; inline audio capped at 20 MB.	**Backup Choice** (drops timestamps honestly).
OpenAI Whisper API	High quality reference implementation.	Requires a paid credit card account.	**Rejected** (breaks free-tier rule).

C. Cloud Hosting Platforms

Platform	Findings & Evaluation	Our Verdict
Render (Free Web Service)	Runs Python and Docker smoothly without credit cards. Allows audio uploads without small file ceilings. Sleeps after 15 min idle.	**Chosen Host** (Live on Render).
Vercel	Has a strict **4.5 MB request body limit**. Real podcast audio files are 20–60 MB, so Vercel blocks them.	**Rejected** (cannot upload audio).
Hugging Face Spaces	Free Docker compute spaces now require paid accounts for new users.	**Rejected** (requires paid plan).

3. How We Keep Answers Trustworthy (Reliability)

Instead of hoping the AI doesn't hallucinate, our Python backend strictly checks everything:

- **Numbered Passages:** The transcript is cut into numbered segments (Passage 1, Passage 2...). The AI must cite these numbers.
- **Verbatim Quote Checks:** For quizzes and Q&A, the server searches the transcript to confirm the quote is real word-for-word. Ungrounded questions are discarded.
- **Honest Timestamps:** Timestamps come only from Whisper or subtitle files (`.srt` / `.vtt`). If plain text is uploaded, the app honestly displays "*No timestamps available*" instead of making up fake times.
- **Server-Side Option Shuffling:** AI models often put the right quiz answer as Option B. We randomize answer positions on the server to prevent bias.
- **Abstaining on Out-of-Domain Questions:** If a user asks something not in the podcast (like "*Who won the World Cup?*"), the system politely refuses instead of guessing.

4. Safety & Security (OWASP Top 10 API Security)

- **File Header Checks (Magic Bytes):** We inspect the binary header of uploaded files. Renaming a text or malicious file to `.mp3` is safely caught and rejected before reaching ffmpeg.
- **Input Guardrails:** Transcripts under 200 characters or over 150,000 characters are safely rejected with clear HTTP error codes.

- **Privacy Notice:** Users are informed upfront that free AI tiers may process their inputs, advising against submitting private passwords or confidential records.
- **Secret Hygiene:** Zero credentials are saved in code or GitHub; all API keys load securely from environment variables.

5. Measured Benchmark: Gemini vs. Groq Inside Our App

We ran both models head-to-head on the exact same 11-minute podcast transcript (about 10,300 characters). Both models were pinned with no hidden fallback:

Benchmark Metric	Google Gemini Flash-Lite	Groq (GPT-OSS-120B)
Overview Generation Time	5.2 seconds	**1.5 seconds** (over 3x faster)
Chapters Generation Time	1.5 seconds	1.8 seconds
Quiz Generation Time	5.8 seconds	**3.5 seconds**
Verbatim Quote Check	**100% Verified** (3 of 3 questions)	**100% Verified** (3 of 3 questions)
Off-Topic Question Refusal	Refused correctly (no guess)	Refused correctly (no guess)
Maximum Episode Length	**Huge** (handles 10+ hour audio)	Medium (limited by 8K tokens/min)

Takeaway: Groq is unmatched for speed (1.5 seconds), while Gemini is unmatched for episode length. Our **Auto mode** uses Groq for speed when possible, and smoothly relies on Gemini for long marathons.

6. Three Future Improvements (Next Steps)

1. **Automated Truthfulness Testing (Highest Priority):** An automated benchmark harness that checks if every summary claim strictly matches the podcast audio without any subtle drift.
2. **Durable Database & Shareable Links:** Storing episode companions in a lightweight database so users can share permanent links with friends.
3. **Word-by-Word Streaming:** Streaming tokens over Server-Sent Events (SSE) so users see summary sentences appear instantly as they are generated.